

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A Case Study

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Q. No	Understanding of Case (25%)	Application of Concepts (25%)	Analysis (20%)	Solution Design (20%)	Clarity (10%)	Total Score (100%)
1						
2						
3						
4						
5						
Total Marks (Out of 100)				Total Marks (Out of 50)		

Annexure A

Case Study

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

1. Hough Transform for Shape Detection

A system uses Hough Transform to detect circular shapes but fails in complex backgrounds.

Analyze the limitations and explain how preprocessing or parameter tuning can improve detection.

2. PCA in Face Recognition

A face recognition system processes high-dimensional image data, resulting in slow performance.

Analyze how PCA helps in reducing dimensionality while preserving important features.

3. Edge Detection in Low Contrast Images

An image has low contrast, causing edge detection algorithms to fail.

Analyze the problem and explain how preprocessing enhances edge detection results.

4. Corner Detection in Textured Regions

A system applies corner detection in highly textured regions, resulting in excessive keypoints.

Analyze the issue and suggest methods to improve feature selection.

5. Feature Matching with Noise

Feature matching between two images fails due to noise in one image.

Analyze the impact of noise and explain how preprocessing improves matching accuracy.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand